Below is a Codex-ready product and technical specification based on your concept. I have converted the initial list of indicators into a clearer application scope, scoring model, data architecture, and initial development plan. The application is intended to assess political, economic, social, demographic, financial, and security risks affecting countries worldwide. Geo Threat

Edit

GeoThreat — Application Build Specification

1. Product Objective

Build a web application called **GeoThreat** that helps citizens, business leaders, investors, analysts, and policymakers identify rising geopolitical threats across the world.

The platform should evaluate risks that may affect:

- National security
- Political stability
- Economic growth
- Business continuity
- Investment decisions
- Supply chains
- Population welfare
- Regional stability

The application should combine long-term structural indicators with recent developments to produce a transparent country-level risk assessment.

The platform must avoid presenting predictions as certainties. It should show:

- The available evidence
- The underlying indicators
- The source and date of each data point
- The level of confidence in the assessment
- The factors driving a country’s risk score

2. Core User Experience

The initial version should allow a user to:

1. View a world map showing geopolitical risk by country.
2. Select a country and open a detailed country profile.
3. Review the country's overall risk score.
4. Review individual risk categories and sub-indicators.
5. Understand why a score has increased or decreased.
6. Compare two or more countries.
7. View historical score changes.
8. Review the sources supporting each assessment.
9. Filter countries by region, risk level, or indicator.
10. identify countries experiencing a material deterioration in risk.

---

3. Target Users

The application should initially support the following user groups:

Citizens and travellers

Users who want to understand political instability, social unrest, access to essential resources, or regional security risks.

Business leaders

Users evaluating market entry, operational continuity, workforce exposure, supply-chain risk, or capital expenditure.

Investors and financial analysts

Users assessing sovereign risk, debt sustainability, economic resilience, demographic trends, and political uncertainty.

Policymakers and researchers

Users comparing countries, identifying structural weaknesses, and analysing regional geopolitical developments.

---

4. Risk Assessment Framework

Each country should receive:

- An overall risk score from 0 to 100
- A risk classification
- A score for each category
- A trend indicator
- A confidence rating
- A list of the strongest risk drivers
- A list of potential stabilising factors

Risk classifications

- **0–19:** Very Low Risk
- **20–39:** Low Risk
- **40–59:** Moderate Risk
- **60–79:** High Risk
- **80–100:** Critical Risk

A higher score represents a higher level of risk.

---

5. Risk Categories

5.1 Political Stability

Assess the durability and legitimacy of the political system.

Indicators may include:

- Government stability
- Frequency of government changes
- Institutional effectiveness
- Rule of law
- Corruption
- Independence of the judiciary
- Electoral integrity

- Concentration of political power
- Restrictions on political opposition
- Frequency of constitutional crises
- Probability of a coup or unconstitutional transfer of power

---

5.2 Political System and Governance

Assess whether the country operates under a democratic, hybrid, authoritarian, or dictatorial system.

Indicators may include:

- Political rights
- Civil liberties
- Freedom of expression
- Freedom of the press
- Independence of public institutions
- Transparency of government decisions
- Accountability of political leaders
- Treatment of opposition parties
- Electoral competitiveness
- Executive constraints

The application should not assume that political system type alone determines national stability. It should evaluate both the nature of the system and its resilience.

5.3 Public Support and Social Unrest

Assess whether the population supports or rejects the country’s political leadership and institutions.

Indicators may include:

- Public approval of national leaders
- Frequency and scale of demonstrations
- Strike activity
- Civil disorder
- Political polarisation

- Trust in government
- Trust in public institutions
- Regional separatism
- Ethnic or religious tensions
- Youth unemployment
- Income inequality
- Cost-of-living pressure
- Online mobilisation and protest activity

The system should distinguish between peaceful democratic protest and violent political unrest.

5.4 Demographic Stability

Assess whether demographic trends support or weaken the country’s long-term stability.

Indicators may include:

- Birth rate
- Death rate
- Fertility rate
- Population growth
- Median age
- Ageing population
- Working-age population
- Dependency ratio
- Life expectancy
- Infant mortality
- Urbanisation
- Population decline
- Regional demographic imbalance

The system should identify whether deaths consistently exceed births and whether demographic decline could affect public finances, labour availability, military capacity, or economic growth.

5.5 Migration and Social Integration

Assess whether the country has a sustainable and effective migration policy.

Indicators may include:

- Net migration
- Labour-force participation among immigrants
- Employment rates among immigrants
- Education outcomes
- Language acquisition
- Citizenship and residency pathways
- Housing pressure
- Social integration
- Contribution to tax revenues
- Use of public services
- Public attitudes toward immigration
- Social or political tensions associated with migration
- Ability to attract skilled workers
- Refugee-processing capacity

The application must avoid discriminatory conclusions. Assessments should be based on measurable policy outcomes and socioeconomic indicators rather than ethnicity, religion, or nationality.

5.6 Regional and External Threats

Assess the extent to which a country is exposed to instability or aggression from neighbouring states or non-state actors.

Indicators may include:

- Armed conflict in neighbouring countries
- Border disputes
- Territorial claims
- Military activity near borders
- Hostile neighbouring governments
- Terrorist organisations
- Proxy warfare
- Sanctions exposure

- Refugee flows
- Regional political instability
- Maritime disputes
- Cyberattacks attributed to foreign actors
- Dependency on insecure trade routes
- Membership in defence alliances

5.7 Access to Essential Resources

Assess whether the population has reliable and affordable access to basic resources.

Indicators may include:

- Drinking-water availability
- Food security
- Electricity access
- Energy affordability
- Heating availability
- Fuel security
- Domestic agricultural capacity
- Energy import dependency
- Food import dependency
- Strategic reserves
- Infrastructure reliability
- Exposure to droughts or extreme weather
- Ability to import essential resources during a crisis
- Diversity of suppliers and import routes

5.8 Education and Human Capital

Assess whether the population has access to an affordable, reliable, and effective education system.

Indicators may include:

- Literacy rate
- School attendance
- Educational attainment
- Teacher availability

- Education expenditure
- University quality
- Vocational training
- Digital skills
- Science and engineering graduates
- Access to international education
- Research output
- Partnerships with leading foreign universities
- Alignment between education and labour-market needs
- Brain drain
- Ability to attract international researchers and students

5.9 Financial and Economic Stability

Assess whether the country can maintain medium- and long-term economic stability.

Indicators may include:

- GDP growth
- GDP per capita
- Inflation
- Unemployment
- Interest rates
- Government budget balance
- Government debt
- Debt-to-GDP ratio
- Foreign-exchange reserves
- Current-account balance
- Currency volatility
- Credit rating
- Access to international capital markets
- Banking-system stability
- Non-performing loans
- Sovereign default risk
- Tax-collection capacity
- Economic diversification
- Exposure to commodity prices
- Foreign direct investment
- Productivity

- Infrastructure investment

The system should assess whether a country can:

- Repay its debts
- Finance essential public services
- Respond to a recession
- Absorb a military, economic, or social shock
- Adapt to technological and industrial change
- Continue participating in the global financial system

5.10 Military and Defence Readiness

Assess whether the country has a modern, credible, and deployable defence capability.

Indicators may include:

- Defence expenditure as a percentage of GDP
- Defence expenditure per capita
- Personnel readiness
- Training levels
- Equipment modernisation
- Logistics capability
- Ammunition and strategic reserves
- Air-defence capacity
- Naval capability
- Cyber-defence capability
- Intelligence capability
- Mobilisation capacity
- Defence-industrial capacity
- Dependency on foreign defence suppliers
- Participation in military alliances
- Public support for the armed forces
- Experience in recent military operations

Military spending alone must not determine the score. The application should consider readiness, sustainability, modernisation, logistics, and strategic relevance.

---

6. Optional Future Categories

The architecture should allow the later addition of:

- Climate and environmental risk
- Public-health resilience
- Cybersecurity risk
- Technology sovereignty
- Critical-infrastructure resilience
- Energy-transition risk
- Supply-chain dependency
- Media freedom and information integrity
- Organised crime
- Terrorism
- Pandemic preparedness
- Space and satellite dependency

---

7. Scoring Methodology

Each category should receive a score between 0 and 100.

Each indicator should contain:

- Current value
- Normalised score
- Weight
- Direction of risk
- Data source
- Publication date
- Last refresh date
- Confidence score
- Optional analyst commentary

Example calculation

```
category_score =
    sum(indicator_normalised_score × indicator_weight)
    / sum(indicator_weight)
```

The overall country score should be calculated as:

```
overall_risk_score =
    sum(category_score × category_weight)
    / sum(category_weight)
```

Initial category weights can be equal. The architecture must allow an administrator to change weights later.

Trend calculation

Each category should also display:

- Improving
- Stable
- Deteriorating
- Rapidly deteriorating

Trend calculations should consider changes over periods such as:

- 30 days
- 90 days
- 12 months
- 5 years

The initial MVP may focus on annual and quarterly data.

---

8. Confidence and Data Quality

Every country score should include a confidence rating:

- High confidence
- Medium confidence
- Low confidence

Confidence should be based on:

- Number of available indicators
- Recency of the data
- Reliability of sources
- Agreement between sources
- Missing-data percentage
- Frequency of data updates

A country with limited or outdated data should not be presented with the same level of certainty as a country with comprehensive and recent information.

---

9. Data Sources

Build the platform so that new sources can be added through connectors or scheduled ingestion jobs.

Potential public sources include:

- World Bank
- International Monetary Fund
- United Nations
- OECD
- World Health Organization
- International Labour Organization
- UNESCO
- UNHCR
- SIPRI
- NATO
- National statistical agencies
- Central banks
- Government finance ministries
- Election authorities
- International energy organisations
- Reputable conflict and political-risk datasets

Every displayed metric must include its source.

Do not use generative AI as the sole factual source. AI may summarise or classify information only when the underlying evidence is retained and visible.

---

10. News and Event Analysis

A later version may ingest trusted news and public reports.

The event-analysis service should:

1. Collect articles from approved sources.
2. Identify the country or countries involved.
3. Classify the event by risk category.
4. Estimate whether it increases or decreases risk.
5. Detect duplicate reporting of the same event.
6. Store source links and publication dates.
7. Produce a concise event summary.
8. Require human review for high-impact score changes.

Examples of events:

- Coup attempt
- Declaration of war
- Government collapse
- Large-scale protest
- Sovereign credit downgrade
- Currency crisis
- Border closure
- Major cyberattack
- Emergency election
- International sanctions
- Natural-resource disruption
- Military mobilisation

---

11. Main Application Pages

11.1 Home Dashboard

Display:

- Global risk map
- Highest-risk countries
- Countries with the fastest deterioration
- Countries showing the strongest improvement
- Recent major geopolitical events
- Regional risk summaries
- Search bar
- Filters

11.2 World Map

Requirements:

- Interactive map
- Countries coloured by risk classification
- Hover state showing country name and overall score
- Click action opening the country profile
- Filters by category
- Filters by region
- Filters by risk level
- Legend explaining the score bands

11.3 Country Profile

Display:

- Country name
- Flag
- Region
- Overall risk score
- Risk classification
- Confidence rating
- Trend
- Last updated date
- Category scores
- Top five risk drivers
- Top five stabilising factors
- Historical trend chart
- Key metrics
- Recent events
- Data sources
- Methodology explanation

11.4 Country Comparison

Allow users to compare between two and five countries.

Display:

- Overall risk score
- Category scores
- Economic indicators
- Demographic indicators
- Political indicators
- Military indicators
- Historical trends
- Confidence levels
- Data completeness

11.5 Methodology Page

Explain:

- How indicators are selected
- How scores are normalised
- How weights are applied
- How missing data is handled
- How trends are calculated
- How confidence levels are determined
- How news events influence scores
- What the application cannot reliably predict

11.6 Administration Interface

Administrators should be able to:

- Add or remove indicators
- Adjust indicator weights
- Adjust category weights
- Add data sources
- Review ingestion failures
- Correct country data
- Approve event-based score changes
- Add analyst commentary
- Publish or unpublish country reports
- View data-quality warnings

---

12. Recommended MVP Scope

The first working version should include:

- Twenty countries
- Ten risk categories
- Five to ten indicators per category
- Manually seeded or CSV-imported data
- Overall and category-level scoring
- Interactive global map
- Country profile page
- Country comparison
- Historical score chart
- Source attribution
- Admin-controlled weights
- Responsive desktop and mobile interface

Do not begin the MVP with automated news ingestion or machine-learning predictions. First establish a reliable, explainable scoring platform.

---

13. Suggested Technology Stack

Use the following stack unless there is a strong technical reason to change it.

Front end

- Next.js
- React
- TypeScript
- Tailwind CSS
- shadcn/ui
- Recharts for charts
- Mapbox GL JS or react-simple-maps for the world map

Back end

- Next.js server actions or API routes for the MVP
- Node.js
- TypeScript

- Zod for validation
- Prisma ORM

Database

- PostgreSQL

Authentication

- Auth.js or Clerk

Data ingestion

- Scheduled Node.js jobs
- CSV import
- REST API connectors
- Background jobs using Inngest, Trigger.dev, or a queue-based worker

Hosting

- Vercel for the application
- Neon, Supabase, or managed PostgreSQL for the database

---

14. Core Data Model

Create the following entities.

Country

```
id
isoCode
name
region
subregion
capital
population
flagUrl
overallRiskScore
riskClassification
confidenceLevel
trend
```

lastUpdatedAt
createdAt
updatedAt

RiskCategory

id
name
slug
description
weight
displayOrder
isActive
createdAt
updatedAt

Indicator

id
categoryId
name
slug
description
unit
weight
minimumValue
maximumValue
higherValueMeansHigherRisk
isActive
createdAt
updatedAt

CountryIndicatorValue

id
countryId
indicatorId
rawValue
normalisedRiskScore
confidenceScore
observationDate
sourceId
notes
createdAt
updatedAt

CountryCategoryScore

`id`

`countryId`

`categoryId`

`score`

`trend`

`confidenceLevel`

`calculationDate`

`createdAt`

`updatedAt`

CountryRiskHistory

`id`

`countryId`

`overallScore`

`riskClassification`

`confidenceLevel`

`recordedAt`

DataSource

`id`

`name`

`publisher`

`url`

`sourceType`

`reliabilityRating`

`updateFrequency`

`lastCheckedAt`

`createdAt`

`updatedAt`

GeopoliticalEvent

`id`

`title`

`summary`

`eventDate`

`countryId`

`categoryId`

`severity`

`riskDirection`

`confidenceScore`

```
status
createdAt
updatedAt
```

EventSource

```
id
eventId
sourceId
articleUrl
publicationDate
```

---

15. API Requirements

Create REST or server-side endpoints for:

```
GET /api/countries
GET /api/countries/:isoCode
GET /api/countries/:isoCode/scores
GET /api/countries/:isoCode/history
GET /api/countries/:isoCode/events
GET /api/categories
GET /api/indicators
GET /api/compare?countries=GBR,FRA,DEU
POST /api/admin/import
POST /api/admin/recalculate
PATCH /api/admin/categories/:id
PATCH /api/admin/indicators/:id
```

Validate all request payloads with Zod.

---

16. Initial Seed Countries

Seed the MVP with a geographically diverse group:

- United Kingdom
- France
- Germany
- United States
- Canada
- Brazil

- China
- India
- Japan
- South Korea
- Russia
- Ukraine
- Poland
- Turkey
- Israel
- Saudi Arabia
- United Arab Emirates
- South Africa
- Nigeria
- Australia

Use sample data clearly labelled as demonstration data until official source ingestion is implemented.

---

17. Design Requirements

The interface should feel:

- Serious
- Analytical
- Neutral
- Trustworthy
- Data-driven
- Easy to understand

Avoid sensationalist language or visual design.

Use:

- Clear typography
- Accessible colour contrast
- Tooltips explaining every score
- Visible data-source references
- Prominent last-updated dates
- Clear warnings where data is incomplete
- Responsive layouts
- Keyboard-accessible controls

Do not rely on red and green alone. Include labels and icons so that users with colour-vision deficiencies can interpret the risk levels.

---

18. Security and Compliance

Implement:

- Role-based access control
- Secure administrator authentication
- Input validation
- Protection against SQL injection
- Rate limiting
- Audit logs for scoring changes
- Secure handling of API keys
- Environment variables for secrets
- No secrets committed to source control
- Privacy notice
- Terms of use
- Methodology disclaimer

The application must not expose private personal data or profile individual citizens.

---

19. Ethical and Editorial Requirements

The platform must:

- Distinguish facts from analysis
- Distinguish structural risk from short-term events
- Avoid presenting disputed territorial claims as settled facts
- Avoid discriminatory conclusions
- Provide source attribution
- Display uncertainty
- Allow data corrections
- Explain scoring changes
- Avoid political advocacy

- Avoid automatically labelling populations as dangerous or unstable
- Require review before publishing extreme or high-impact conclusions

---

20. Initial Codex Implementation Task

Create a production-quality MVP repository for **GeoThreat**.

Required first implementation

1. Initialise a Next.js application using TypeScript.
2. Configure Tailwind CSS.
3. Add shadcn/ui components.
4. Configure PostgreSQL and Prisma.
5. Implement the database schema described above.
6. Add seed data for the twenty initial countries.
7. Add ten risk categories.
8. Add sample indicators and demonstration scores.
9. Build the global dashboard.
10. Build an interactive country risk map.
11. Build the country profile page.
12. Build the country-comparison page.
13. Build the methodology page.
14. Add an administrator page for editing weights.
15. Implement the score-calculation service.
16. Add unit tests for the score calculations.
17. Add API validation with Zod.
18. Add loading, empty, and error states.
19. Add a README with local setup instructions.
20. Add an `.env.example` file.

Code-quality requirements

- Use strict TypeScript.
- Avoid `any`.
- Use reusable components.
- Separate UI, domain logic, data access, and scoring logic.
- Add comments only where logic is not self-explanatory.

- Use descriptive names.
- Include database migrations.
- Include automated tests.
- Include linting and formatting.
- Ensure the project can run locally without proprietary services.
- Use mock or seed data where external API credentials are unavailable.

Initial repository structure

```
src/
  app/
    page.tsx
    countries/
      [isoCode]/
        page.tsx
    compare/
      page.tsx
    methodology/
      page.tsx
    admin/
      page.tsx
    api/
  components/
    dashboard/
    map/
    country/
    comparison/
    charts/
    layout/
    ui/
  domain/
    scoring/
    countries/
    indicators/
    events/
  lib/
    db/
    validation/
    utils/
  services/
    scoring-service.ts
```

```
    country-service.ts
    data-source-service.ts
prisma/
  schema.prisma
  seed.ts
tests/
```

---

21. Codex Execution Instructions

Start by producing:

1. The proposed folder structure.
2. The Prisma schema.
3. The scoring-engine implementation.
4. The seed-data structure.
5. The dashboard page.
6. The country-profile page.
7. The setup README.

Then implement the remaining features incrementally.

Before completing the task:

- Run the TypeScript compiler.
- Run the linter.
- Run all tests.
- Resolve compilation and test failures.
- Confirm the application starts locally.
- Summarise the files created.
- Identify any features that still use demonstration data.

This version is ready to paste into Codex as the primary build brief.